

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (E&E) (Credit System) Degree Examinations**

May - June 2023

20EE604 - VLSI CIRCUITS AND DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Colour pencils / pens can be used for stick diagrams.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Derive the expression for drain current in linear and saturation region for nMOS transistor.	10	L*3	2	1,2,5,12
	b) Explain step by step CMOS N-Well fabrication process.	10	L2	1	1
2.	a) Derive the small signal AC transfer characteristic model of a n type MOSFET.	10	L3	2	1,2,5,12
	b) Explain the following second order effects in a MOSFET. (i) Channel Length Modulation (ii) Hot carrier Effects	10	L2	2	1,2,5,12
3.	a) Derive the expression for threshold voltage in an enhancement nMOS transistor and hence solve. Consider a MOS system with the following parameters: $N_A=3 \times 10^{16} \text{ cm}^{-3}$, $t_{ox}=200 \text{ \AA}$, $\epsilon_{ox}=3.9 \times 8.85 \times 10^{-14} \text{ F/cm}$, $\epsilon_{Si}=11.7 \times 8.85 \times 10^{-14} \text{ F/cm}$ and $q=1.6 \times 10^{-19} \text{ Coulomb}$ Determine the substrate-bias (or body-effect) coefficient Determine surface potential with body at zero potential Determine the change in threshold voltage ($V_T - V_{T0}$) if $V_{SB}=2.5 \text{ V}$	10	L3	2	1,2,5,12
	b) Explain the following: (i) Moore's law in VLSI. (ii) Process of wafer fabrication by Czochralski method.	10	L2	1	1
Unit – II					
4.	a) Define stick diagram and explain the rules and hence draw the stick diagram for $y = (A + B)C$ using CMOS logic.	10	L3	3	1,2,5,12
	b) Explain the following MOSFET Scaling methods. (i) Constant Field Scaling (ii) Constant Voltage Scaling	10	L2	3	1,2,5,12
5.	a) With the help of sheet resistance model of a MOS transistor, calculate the channel resistance of MOS transistor of size as shown in Fig. Q 5(a).				

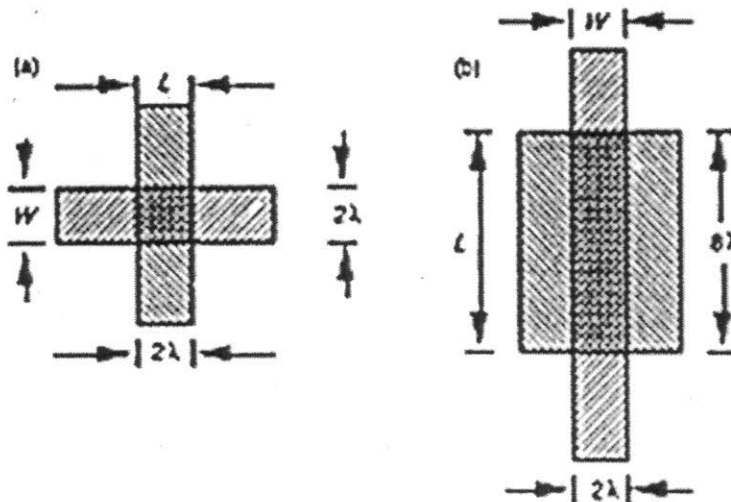


Fig. Q 5(a)

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|-------|---|----|----|---|----------|
| b) | Explain the general circuit structure of a resistive load nMOS inverter and its voltage transfer characteristics. | 10 | L2 | 4 | 1,2,5,12 |
| 6. a) | Explain CMOS inverter delays and hence estimate the rise time and fall time. | 10 | L3 | 3 | 1,2,5,12 |
| b) | Explain the CMOS inverter. | | | | |
| | (i) Operating regions | | | | |
| | (ii) Static characteristics | 10 | L2 | 4 | 1,2,5,12 |

Unit – III

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|-------|--|----|----|---|----------|
| 7. a) | Realize the Two input NAND gate using nMOS technology and draw the stick diagram and then derive the boundary condition. | 10 | L3 | 5 | 1,2,5,12 |
| b) | Explain a 4 bit dynamic shift register using nMOS and CMOS logic along with the stick diagram of shift register cell. | 10 | L2 | 5 | 1,2,5,12 |
| 8. a) | Realize the Two input NOR gate using CMOS technology and draw the stick diagram. | 10 | L3 | 5 | 1,2,5,12 |
| b) | Explain a parity generator circuit and hence obtain the stick diagrams using (i) nMOS logic (ii) CMOS logic | 10 | L2 | 5 | 1,2,5,12 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
